

AEFLES

Adaptive Fiber Load Estimation System

Business Concept: Fiber Optic Structural Health Monitoring for Aerospace

Mission

Modern aerospace systems operate under extreme mechanical stress, where structural failure can result in catastrophic consequences. Ensuring real-time awareness of structural integrity is critical, yet many current sensing methods rely on bulky, expensive, or limited-point measurement systems that fail to provide continuous insight across an entire structure. The inspiration for AEFLES comes from the need for a lightweight, scalable, and precise sensing solution that can be integrated directly into aerospace structures. Fiber optic sensing offers a promising alternative, enabling continuous monitoring of strain and load distribution across wings, fuselages, and other critical components without adding significant weight or complexity.

Value Proposition

The core problem AEFLES addresses is the lack of efficient, real-time structural monitoring in aerospace systems. Traditional sensors are often limited to discrete points, require complex wiring, and increase system weight and cost.

AEFLES introduces a fiber optic-based sensing system that measures structural strain and load by detecting changes in light transmission through optical fibers embedded or attached to components. This enables:

- Continuous, real-time monitoring across large surfaces
- Reduced system weight compared to traditional sensors
- Increased reliability through fewer electrical components
- Early detection of structural stress and potential failure

This solution improves flight safety, reduces maintenance costs, and enables predictive diagnostics, allowing engineers to identify issues before they become critical.

Market Opportunity

The demand for structural health monitoring is growing across aerospace, defense, and unmanned aerial systems. Aircraft manufacturers, research institutions, and drone developers require accurate, lightweight, and scalable sensing solutions to improve performance and safety.

Current solutions rely heavily on strain gauges and electronic sensors, which are limited in coverage and scalability. AEFLES positions itself as a next-generation sensing system that leverages fiber optics for enhanced performance.

As aerospace systems become more advanced and data-driven, the need for integrated sensing technologies will continue to expand, creating strong long-term market potential.

Customer Discovery and Segments

Primary customer segments include:

- Aerospace manufacturers and suppliers
- UAV and drone developers
- Research institutions and engineering labs
- Defense and government aerospace programs
- University engineering programs

Customer discovery will begin through discussions with engineering students, faculty, and aerospace enthusiasts, as well as outreach to UAV developers and research labs. Early validation will focus on:

- Interest in lightweight sensing solutions
- Current limitations of existing monitoring systems
- Willingness to adopt lower-cost, scalable alternatives

Product–Market Fit

AEFLES achieves product–market fit by addressing key challenges in aerospace systems:

- Need for real-time structural monitoring
- Demand for lightweight and non-intrusive sensors
- Desire to reduce maintenance costs and downtime

Validation will occur through prototype testing and feedback from early users. Demonstrating that AEFLES can accurately detect strain and load changes in controlled environments will establish credibility as a viable alternative to traditional sensing methods.

Competitive Advantage

The current market is dominated by traditional strain gauges and electronic sensing systems. These solutions:

- Provide limited measurement coverage

- Add weight and complexity
- Require extensive wiring and maintenance

AEFLES offers a competitive advantage through:

- Fiber optic sensing technology enabling distributed measurements
- Lightweight integration into existing structures
- Scalability across large surfaces and complex geometries
- Reduced electrical dependency, improving reliability

There is also potential for intellectual property in signal processing methods, system integration techniques, and low-cost implementation of fiber optic sensing systems.

Conclusion

AEFLES introduces an innovative approach to structural monitoring by leveraging fiber optic sensing to deliver real-time, lightweight, and scalable load estimation. By improving safety, reducing maintenance costs, and enabling predictive analysis, AEFLES has the potential to modernize how aerospace systems are monitored and maintained.